THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** tissue	Tissue.
2	**(d)** sclerenchyma	Dead, with lignin.
3	**(b)** epithelial tissue	Covering.
4	**(d)** striated (voluntary) muscles	Skeletal.
5	**(a)** meristematic tissue	Dividing cells.
6	**(c)** xylem	Tracheids and vessels.
7	**(c)** squamous epithelium	Flat cells.
8	**(a)** smooth (involuntary) muscles	Unstriated.
9	**(b)** intercalary meristem	At the base of leaves/internodes.
10	**(c)** phloem	Sieve tubes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Group of similar cells doing a specific function (1); Division of labour makes bodies efficient (1).	1+1
12	Xylem carries water and minerals (1); Phloem carries food (1).	1+1
13	Protection (1); Absorption and secretion (1).	1+1
14	Meristematic: dividing cells, small vacuoles (1); Permanent: do not divide, fixed function (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Grow onion roots in water (1); Measure the length daily (1); Growth takes place near the tip due to apical meristem (1).	1+1+1
16	Cell body with nucleus (1); Dendrites receive impulses (1); Axon carries the impulse away (1).	1+1+1
17	Voluntary: controlled by will, e.g. biceps (1); Involuntary: not controlled, e.g. heart, intestine (1); Different tissues (1).	1+1+1
	OR: Leaf epidermis: stomata, cuticle (1); Root epidermis: root hairs absorb water (1); Protects underlying tissues (1).	1+1+1
18	Squamous: skin, alveoli, protection/exchange (1); Columnar: intestine, absorption (1); Cuboidal: kidney tubules, secretion (1).	1+1+1
	OR: A hard substance in the cell wall (1); Found in sclerenchyma and xylem (1); Gives strength and makes the cells waterproof (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Apical: tips, growth in length (1); Lateral: girth (1); Intercalary: base of leaves/internodes (1); Cells dividing, thin walls (1); Differentiate to form permanent tissues (1).	1+1+1+1+1
	OR: Striated (1); Smooth (1); Cardiac (1); Voluntary or involuntary (1); Locations (1).	1+1+1+1+1
20	Parenchyma (1); Chlorenchyma and aerenchyma (1); Collenchyma (1); Sclerenchyma (1); Locations and functions (1).	1+1+1+1+1
	OR: Neuron structure (1); Dendrites to cell body to axon (1); Nerve endings (1); Role in coordination (1); Nerve and nervous system (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that xylem conducts food and phloem conducts water; xylem: water and minerals, phloem: food.
- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).
- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.

- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** columnar	Absorption.
2	**(d)** cardiac muscles	Involuntary, striated and branched.
3	**(a)** lateral meristem (cambium)	Secondary growth.
4	**(a)** xylem parenchyma	Others are dead.
5	**(d)** cuboidal	Absorption and secretion.
6	**(b)** neuron	Nerve cell.
7	**(d)** tips of roots and shoots	Increases length.
8	**(c)** sieve tubes, companion cells, phloem fibres and phloem parenchyma	Phloem.
9	**(b)** stratified squamous epithelium	Protection.
10	**(b)** dendrites, cell body and axon	Structure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Exchange of gases (1); Loss of water vapour (transpiration) (1).	1+1
12	Tendon joins muscle to bone (1); Ligament joins bone to bone (1).	1+1
13	They are actively dividing (1); Dense cytoplasm and prominent nucleus are needed (1).	1+1
14	Striated (skeletal) (1); Smooth and cardiac (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tracheids, vessels, xylem parenchyma, fibres (1); Most are dead with lignified walls (1); Conduct water and minerals upward and give support (1).	1+1+1
16	Blood: transports materials (1); Bone: support and protection (1); Adipose: fat storage and insulation (1).	1+1+1
17	Growing tissue in a sterile nutrient medium (1); Growing identical plants (1); Producing disease-free plants (1).	1+1+1
	OR: Sieve tubes with sieve plates (1); Companion cells, parenchyma, fibres (1); Transport food from leaves (1).	1+1+1
18	Intercalary meristem at the base of leaves (1); Continues to divide (1); New growth from the base (1).	1+1+1
	OR: Muscle cells contract (1); Nerve cells carry messages over long distances (1); Structure suits function (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tracheids and vessels (1); Xylem parenchyma and fibres (1); Sieve tubes (1); Companion cells (1); Functions (1).	1+1+1+1+1
	OR: Bones support (1); Muscles pull (1); Tendons connect (1); Joints allow movement (1); Ligaments hold (1).	1+1+1+1+1
20	Epidermis (1); Cuticle (1); Stomata (1); Root hairs (1); Cork (1).	1+1+1+1+1
	OR: Single carrot phloem cells (1); Grown in medium with coconut milk (1); Developed into plants (1); Totipotency (1); Applications: cloning, disease-free plants (1).	1+1+1+1+1

Mistakes to avoid:

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- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).
- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.
- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** epidermis	Often with cuticle.
2	**(d)** guard cells	Control opening.
3	**(b)** respiratory tract	Moves mucus.
4	**(c)** pivot joint	Allows rotation.
5	**(d)** cork cambium	Protective tissue in older stems.
6	**(d)** exchange of gases and transpiration	Gas exchange.
7	**(a)** connective tissue	Connective.
8	**(b)** hinge joints	Move in one plane.
9	**(c)** aerenchyma	Air-filled.
10	**(d)** cuticle	Reduces water loss.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Epidermis is replaced by cork as the stem thickens (1); Cork is protective, impermeable (1).	1+1
12	Parenchyma: thin walls, storage (1); Collenchyma: thickened corners, flexibility (1).	1+1
13	Bone is hard with calcium salts (1); Cartilage is flexible and softer (1).	1+1
14	Receives and transmits nerve impulses (1); Helps control and coordination (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parenchyma: living, thin wall, storage (1); Collenchyma: living, thick corners, flexibility (1); Sclerenchyma: dead, lignin, strength (1).	1+1+1
16	Hinge: knee (1); Ball and socket: shoulder (1); Pivot: neck; fixed: skull (1).	1+1+1
17	Epidermis (1); Cortex and vascular bundles (1); Pith (1).	1+1+1
	OR: Striated: voluntary, cylindrical (1); Smooth: involuntary, spindle (1); Cardiac: involuntary, branched, striated (1).	1+1+1
18	Binds organs (1); Gives support (1); Stores fat, transports materials (1).	1+1+1
	OR: Plant tissues mostly give support, animals' tissues organise movement (1); Plant growth is localised in meristems (1); Animals have more specialised tissues (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Squamous (1); Cuboidal (1); Columnar (1); Ciliated (1); Glandular (1).	1+1+1+1+1
	OR: Growth (1); Organisation (1); Meristem vs dividing cells everywhere (1); Supporting tissues (1); Conducting tissues (1).	1+1+1+1+1
20	Blood (1); Bone (1); Cartilage (1); Ligament and tendon (1); Areolar and adipose (1).	1+1+1+1+1
	OR: Xylem: hollow, lignified (1); Phloem: sieve plates (1); Epidermis: tightly packed (1); Root hairs: long to absorb (1); Structure fits function (1).	1+1+1+1+1

Mistakes to avoid:

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